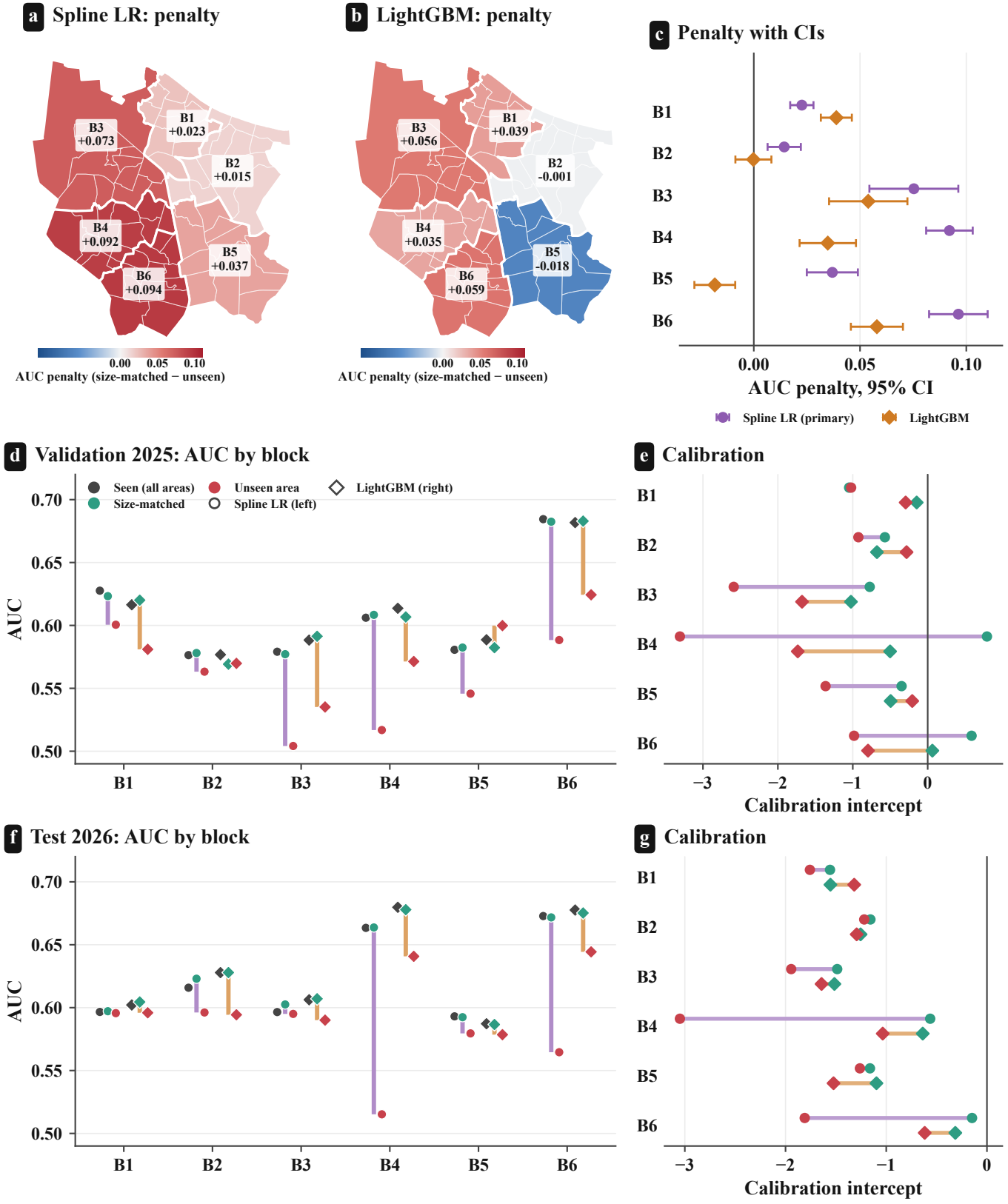

a Spline LR: penalty
B3 +0.073
B1 +0.023
B2 +0.015
B4 +0.092
B5 +0.037
B6 +0.094
0.00 0.05 0.10
AUC penalty (size-matched − unseen)
b LightGBM: penalty
B3 +0.056
B1 +0.039
B2 -0.001
B4 +0.035
B5 -0.018
B6 +0.059
0.00 0.05 0.10
AUC penalty (size-matched − unseen)
c Penalty with CIs
B1
B2
B3
B4
B5
B6
0.00 0.05 0.10
AUC penalty, 95% CI
Spline LR (primary)
LightGBM
d Validation 2025: AUC by block
Seen (all areas)
Size-matched
Unseen area
Spline LR (left)
LightGBM (right)
AUC
0.70 0.65 0.60 0.55 0.50
B1 B2 B3 B4 B5 B6
e Calibration
B1 B2 B3 B4 B5 B6
−3 −2 −1 0
Calibration intercept
f Test 2026: AUC by block
AUC
0.70 0.65 0.60 0.55 0.50
B1 B2 B3 B4 B5 B6
g Calibration
B1 B2 B3 B4 B5 B6
−3 −2 −1 0
Calibration intercept